📖 Rules

As a team you plan Germany's electricity up to 2045 – in **four stages**. In each stage you build new plants and decide about old power plants. Then the computer calculates a whole year, hour by hour. You get points for three goals: ⚡ **always enough power**, 🌿 **climate protection** (clean power and little CO_2) and 💶 **as cheap as possible** – up to 5 points each per stage, at most 60 in total.

In short: cover demand in every hour, make the power clean, watch the costs – and plan a buffer for what you cannot foresee.

1 How a stage works

1. Build

Spend build tokens on new plants, keep or retire old power plants.

2. Test

Try it out with the simulator (3× per stage). It calculates without chance.

3. Mark as ready

You can still make changes until the game master closes the build phase. After that it counts.

4. Simulate the year

The same for everyone: weather, demand, maybe a disaster.

5. Points

Look at the results – then on to the next stage.

2 Your screen

Build
Build plants and keep old power plants. On the right, "Your power plant fleet": is the firm capacity above the peak load?

Influence
Use influence tokens for measures such as hydrogen or acceptance.

Simulator
Test run of your plan with a chart.

Results
What really happened in the year – with points and charts.

History
Your points and decisions in all stages.

Header
Build tokens as boxes (filled = planned, orange = credit) and your points.

3 Goals & points

Each stage gives up to 5 points per goal. The goals get stricter from stage to stage.

⚡ Security of supply

Hours per year in which power is missing. Fewer is better.

At most … hours:

Stage	5 pts	4	3	2	1
all	3	10	25	50	100

🌿 Climate protection

Two conditions – the worse value counts. (1) Share of CO_2-free power: what is generated cleanly in Germany; exported green power counts, imports do not. (2) CO_2 emissions in Germany (Mt per year) – the limit falls every stage to almost zero (climate neutrality 2045). 5 points = political target reached.

Share CO_2-free, at least … %:

Stage	5 pts	4	3	2	1
▶ 1	80	75	70	65	60
2	88	84	80	76	72
3	93	90	87	84	80
4	97	95	92	89	85

CO_2 emissions, at most … Mt:

Stage	5 pts	4	3	2	1
▶ 1	70	80	90	105	120
2	45	55	65	80	95
3	25	35	50	65	80
4	8	20	35	50	65

Orientation: 2030 Climate Protection Act (energy sector 108 Mt, of which electricity roughly 70), 2035 G7 pledge “power sector largely CO_2-free”, 2040 coal exit and end of new CO_2 allowances in the EU emissions trading system, 2045 climate neutrality (waste incineration alone ~5 Mt).

💶 Affordability

Costs of plants, fuel and CO_2 per MWh of electricity (excl. grid, taxes, levies). Investments are spread over their lifetime and also count in all later stages. Lower is better.

At most … €/MWh:

Stage	5 pts	4	3	2	1
▶ 1	58	63	68	73	79
2	64	69	74	80	86
3	63	68	74	80	87
4	65	70	76	83	91

Up to 15 points per stage, 60 in the whole game. In a tie, the team with less CO_2 wins.

4 The four stages

Demand grows because of heat pumps, electric cars and industry.

Stage 1
Now–2030

Demand	**599 TWh**
Peak load	**106 GW**
Build tokens	**18**
CO_2 price	**134 €/t**
CO_2 target	**≤ 70 Mt**

⚠️ Coal at most 17 GW. Use influence tokens early.

Stage 2
2031–2035

Demand	**809 TWh**
Peak load	**143 GW**
Build tokens	**18**
CO_2 price	**154 €/t**
CO_2 target	**≤ 45 Mt**

⚠️ Big jump: demand +35 %, peak load +37 GW – and only 6 GW of coal left.

Stage 3
2036–2040

Demand	**928 TWh**
Peak load	**163 GW**
Build tokens	**17**
CO_2 price	**173 €/t**
CO_2 target	**≤ 25 Mt**

⚠️ Coal and oil are gone, reserve only 4 GW. Many old wind turbines go offline.

Stage 4
2041–2045

Demand	**980 TWh**
Peak load	**166 GW**
Build tokens	**17**
CO_2 price	**186 €/t**
CO_2 target	**≤ 8 Mt**

⚠️ Climate-neutral: 5 climate points only with ≥ 97.0 % CO_2-free and ≤ 8 Mt CO_2.

Demand incl. grid losses, excl. electrolysis. Peak load in a normal weather year – in cold winters it is higher.

5 Tokens

Tokens	What it is
Build tokens	€10 bn for new plants. Per stage: 18 / 18 / 17 / 17 – more at the beginning, because that is where most has to be built. **They expire at the end of the stage** – so use them up.
Credit tokens	Up to 4 extra build tokens per stage, they cost 20 % more.
Influence tokens	2 at the start, more for 2 build tokens each. They unlock a measure – each only once per game. The earlier, the longer it works.

Measure	Effect
Policy & society	
Acceptance & participation	Halves the risk of public backlash – from this stage.
Accelerated innovation	Double cost reduction from your own construction projects – from this stage.
Climate resilience	Halves the impact of disasters (including dark doldrums, heat) – from this stage.
Technology & infrastructure	
Hydrogen demonstration programme	Your own green hydrogen: electrolysis, storage and conversion of gas turbines – from the next stage. Electrolysis is only available with this card. It runs only on German surplus green power.
Hydrogen grid & imports	Imported hydrogen for converted gas turbines, plus storage and conversion (no electrolysis) – from the next stage. Expensive, but CO_2-free. Imports: 2035: 15 GW H_2 (≈ 6 GW electricity) at 140 €/MWh H_2; 2040: 30 GW H_2 (≈ 12 GW electricity) at 125 €/MWh H_2; 2045: 45 GW H_2 (≈ 18 GW electricity) at 110 €/MWh H_2.
European grid expansion	+10 GW of interconnectors to other countries for imports and exports – from the next stage.
Flexible demand	Electric cars, heat pumps and industry adapt their consumption – from the next stage. Consumption is shifted within a day; industry cuts back when power is scarce (from 300 €/MWh). Grows with every stage.

6 New plants

Values for Now–2030. 1 build token = €10 bn.

Plant	per token	existing	Capacity factor	Protest risk	What for?
Onshore wind	+7.2 GW	57.3 GW	26 %	medium	Cheap power when the wind blows; delivers more in winter than in summer.

Offshore wind	+3.6 GW	10.6 GW	45 %	low	Blows more steadily and strongly than on land, but more expensive.
Ground-mounted PV	+20.7 GW	37.6 GW	11 %	medium	Cheapest power, but only during the day and mainly in summer.
Rooftop PV	+11.3 GW	73.0 GW	10 %	none	Like ground-mounted PV, ~10 % less yield, more expensive – but hardly any opposition.
Battery storage (4-hour)	+10.3 GW	0.0 GW	Storage (4 h full load)	none	Shifts power, e.g. from noon to the evening; when full it lasts 4 h at full power.
Hydrogen: electrolysis	+5.3 GW	0.0 GW	when there is a surplus	low	Electrolysis: turns German surplus green power into hydrogen – for storage and converted gas turbines.
Hydrogen storage (cavern)	+3.7 TWh H_2	0.0 TWh H_2	weeks–months	low	Salt cavern: keeps hydrogen for weeks and months – cheap per TWh.
New gas turbines (H_2-ready)	+15.6 GW	0.0 GW	dispatchable	medium	Cheap gas turbines as a reserve: ready to start at any time, emit CO_2 – can be converted to H_2.
Biomass flexibilisation	+8.3 GW	0.0 GW	dispatchable	low	More capacity at biogas plants: same amount of energy, but targeted at scarce hours.

- **Hydrogen needs three building blocks: electrolysis** makes hydrogen from German surplus green power (not from imported, coal or gas power), the **storage** (salt cavern) keeps it for weeks, **converted gas turbines** turn it back into electricity (only about a quarter comes back). Two influence cards, each from the following stage: “Hydrogen demonstration programme” = your own green hydrogen (electrolysis, storage, conversion); “Hydrogen grid & imports” = expensive hydrogen from abroad (storage, conversion, imports, but no electrolysis). Both together cost two tokens. Import capacity is given in GW of hydrogen – only about 40 % of it becomes electricity. The game shows how many days your storage lasts.
- **Biomass flexibilisation:** the same amount of biogas, but more capacity exactly when power is scarce (at most three times the biomass you keep).
- **Old wind and solar plants** go offline step by step. The cards in the Build tab show how much is still running in the next stage.
- **Learning effect:** what you build gets a little cheaper for you in the following stages.

7 Old power plants & fixed plants

You can keep or retire old power plants – retiring is final, keeping them costs operating costs.

Power plant	today	Protest risk	Note
Coal (lignite & hard coal)	21.9 GW	high	always runs at least at 15 % (district heating, mines) and changes its output by at most 40 % per hour. Coal phase-out by law: at most 17 GW (2030), 6 GW (2035), 0 GW (2040), 0 GW (2045)
Gas power plants	30.1 GW	low	dispatchable, emit CO_2
Biomass/biogas	8.9 GW	low	dispatchable, but only ~41 TWh per year

Fixed plants – every team has them, they cannot be changed. They follow a fixed schedule: old plants go offline, the interconnectors to other countries are expanded automatically.

Plant	2030	2035	2040	2045	Note
Pumped hydro	9.9 GW	9.9 GW	9.9 GW	9.9 GW	Energy content lasts about 6 hours at full power

Old large batteries	3.7 GW	3.4 GW	1.5 GW	0.0 GW	Energy content for about 1 hour at full power; retired by 2045
Run-of-river hydro	3.9 GW	3.9 GW	3.9 GW	3.9 GW	small, fairly steady, CO_2-free
Waste incineration	1.9 GW	1.9 GW	1.9 GW	1.9 GW	always running, half renewable
Oil & other	4.8 GW	4.8 GW	0.0 GW	0.0 GW	expensive, only when scarce
Reserve power plants	11.1 GW	11.1 GW	3.6 GW	3.6 GW	only in emergencies, fossil
Interconnectors to other countries	25.0 GW	28.0 GW	32.0 GW	35.0 GW	at most this much import or export; +10 GW with "European grid expansion" (from the following stage)

8 Chance & risks

Weather, demand and disasters are the same for all teams. Learning effects and protests hit each team individually.

- **Weather:** usually a normal year (2008), sometimes a difficult one (1995: less wind), rarely an extreme winter (2009: long cold spell, hardly any wind).
- **Demand:** can be a few percent higher or lower than expected.
- **Disasters** become more likely from stage to stage: *storm/flood* – dice are rolled for each technology separately, those hit deliver only half for 4 weeks. *Cold dark doldrums* – 2 weeks with hardly any wind and sun. *Heat & drought* – power plants with cooling water (gas, coal, biomass) deliver less, run-of-river too. Wind, sun and storage are not affected.
- **Neighbouring countries** help out with power – but not when there is a lull across all of Europe at the same time. They partly take surplus power (e.g. for their hydro storage) and give it back later at a higher price. Exported green power counts for the CO_2-free share, imported power does not. For CO_2 emissions only what is emitted in Germany counts – exports do not help there.
- **Protest (backlash):** every build token can trigger protest. *New plant:* what was built keeps running, but in the next stage you may not build more of it. *Old power plant:* must be retired completely in the next stage.
- **Last stage:** every protest against a new plant costs 2 points – so that nobody builds everything without a second thought at the end.

9 Tips for the start

1. **Test first, then mark as ready.** The simulator knows no chance – plan a reserve.
2. **Use up your build tokens** – leftovers expire.
3. **Wind and sun alone are not enough.** On windless winter evenings there must still be power. Keep the firm capacity above the peak load.
4. **Batteries help for hours, not for weeks.** For long lulls you need gas, biomass or hydrogen.
5. **Think about hydrogen early:** it only takes effect one stage after the influence token. Your own hydrogen needs three building blocks (electrolysis, storage, converted gas turbines), imports need the hydrogen grid. Without hydrogen, 2045 will hardly be climate-neutral.
6. **Stage 2 is the big jump:** much more demand, much less coal – prepare already in stage 1.
7. **Cut CO_2 stage by stage:** the emission limit falls every stage (at the top of the overview: "CO_2 target"). Gas power plants run whenever they are needed – less CO_2 only comes with more green power, storage, hydrogen conversion or by retiring old power plants.

10 How does the model calculate?

Show for those interested

- Germany as one grid without bottlenecks, all **8,760 hours** of a year. The neighbouring countries are simplified as one market, with their own weather and demand, a storage (hydropower in Norway and the Alps) and a fixed uptake for German surplus power.
- In every hour the **cheapest** plants run first. Storage charges when there is a surplus and delivers when power is scarce. The model knows the whole year in advance.
- **Costs** = investment (spread over the lifetime, 5 % interest) + operating costs + fuel and CO_2 + imports − export revenue + credit and influence costs, divided by demand. Old plants only count with their operating costs. Missing power is shown as damage (10,000 €/MWh) but only costs supply points, not cost points.
- **CO_2-free** = (demand + exports − imports − fossil power) ÷ demand, at most 100 %. CO_2-free are wind, sun, hydro, biomass, hydrogen and half of waste incineration.
- **CO_2 emissions** only count in Germany, as in the Climate Protection Act. The CO_2 of imports is shown separately.
- **Climate points** = the worse of the points for the share and for the emissions. So exports do not help against too much CO_2, and imports do not help against too small a share.
- Data: ENTSO-E TYNDP 2024 (weather, demand, prices), Federal Network Agency (power plants), PyPSA technology-data (costs).

11 Glossary – terms explained simply

Term	Meaning
Capacity (GW)	How much power can flow at one moment – how far a tap is turned on. 1 gigawatt (GW) is roughly one large power plant.
Energy (TWh)	How much power flows in total over a period – how much water ends up in the bucket. Capacity times hours. Germany uses about 500 terawatt hours (TWh) per year today.
Peak load	The hour with the highest demand of the year, usually on a cold winter evening. That much capacity must be available then.
Capacity factor	How much a plant delivers on average over the year, compared with full output around the clock. An onshore wind turbine manages about a quarter, a solar plant about a tenth.
Dispatchable	Delivers power when it is needed – e.g. gas, biomass, hydrogen or a full storage. Wind and sun are not dispatchable: they only deliver when the weather plays along.
Firm capacity	All dispatchable power plants together (gas, hydrogen, coal, biomass, reserve). You can count on them even when the wind does not blow and the sun does not shine for days. It should be above the peak load. Storage and flexible demand also help – but only for a few hours. Hydrogen turbines only count as far as the hydrogen lasts for 2 weeks.
Residual load	Demand minus wind and solar power – the rest that dispatchable plants, storage and imports have to supply.
Dark doldrums (Dunkelflaute)	Several days with little wind and hardly any sun, usually in winter. The toughest test for the power system.
Supply gap	An hour in which there is not enough power. In reality, consumers would then have to be disconnected.
Storage	Takes in power when there is too much and releases it when it is missing. “4 h” means: the full storage lasts 4 hours at full power (8 hours at half power). Batteries and pumped hydro therefore last for hours, hydrogen for weeks.
Hydrogen & electrolysis	Electrolysis turns electricity into hydrogen. It can be stored in underground salt caverns for weeks and months and burned again without CO_2 in converted gas turbines. A lot is lost on the way: only about a quarter of the electricity comes back. Electrolysis is expensive, the storage is comparatively cheap per TWh. Hydrogen can also be imported via pipelines (hydrogen core grid).

Curtailment	Wind and solar power that can currently neither be used, stored nor exported. The plants are then throttled – the power is lost.
Imports & exports	Via power lines to other countries (interconnectors) Germany can buy (import) or sell (export) electricity.
Flexible demand	Consumers that think along: an electric car charges at noon instead of in the evening, a heat pump preheats, a factory pauses in expensive hours. This lowers the peak load.
CO_2 price	What power plants have to pay for every tonne of CO_2 they emit. Makes coal and gas more expensive – and rises until 2045.
CO_2 emissions (Mt)	How much CO_2 the power plants in Germany emit in one year, in million tonnes (Mt). For comparison: under the Climate Protection Act the whole energy sector may still emit 108 Mt in 2030.
Climate neutrality (net zero)	Target of the Climate Protection Act for 2045: Germany emits no more greenhouse gases than forests, moors and technology absorb again. For the power sector this means: emissions close to zero. Tonnes are counted – green power exports do not offset anything.
CO_2-free share	How much of the demand in Germany is generated cleanly (wind, sun, hydro, biomass, hydrogen). Exported green power counts, imports do not. Legal target for 2030: 80 %.

Game idea based on “Path to Zero” by Jesse D. Jenkins and the Princeton ZERO Lab · Free software (GPL-2.0-or-later) · Learning game, not a forecast